

ECE410 Study Sheet — RBF-SVM ASIC (m5/v10)

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Key Project Numbers (v10 / m5 Final)

Technology	sky130_fd_sc_hd	Feature dims	256 (128+64+64)
Clock	40 MHz (25 ns)	SV allocation	[95,95,95,120,95]=500
Cell count	157,991	Q6.10 accuracy	98.33% (295/300)
Core utilization	15.0%	Quant. flips	0
TT WNS	+3.96 ns (0 viol)	LAT / inference	3 / 9.87 ms
Active power	55.25 mW	Classes	Normal,PVC,AFib,VT,SVT
Avg power @ 80 bpm	0.869 mW	γ	0.25 $\rightarrow$ 0x0100

Fixed-Point Arithmetic ($\mathbb{Q}_{6.10}$)

A 16-bit signed value: $\hat{v} = \lfloor v \cdot 2^{10} \rfloor \in [-32768, 32767]$, $v = \hat{v}/1024$, $\text{LSB} = 2^{-10} \approx 0.001$.

Multiplication of two $\mathbb{Q}_{6.10}$ values produces a raw 32-bit $\mathbb{Q}_{12.20}$ product (integer bits 6+6 = 12; fractional bits 10+10 = 20). Arithmetic right-shift by 10 restores $\mathbb{Q}_{6.10}$:

$$\widehat{a \cdot b} = \frac{\hat{a} \cdot \hat{b}}{2^{10}} \quad (\text{truncate to 16-bit after shift})$$

Shift is arithmetic (sign-preserving); truncation discards only redundant sign-extension bits.

LUT maximum index: $D_{\max} = 256 \times 64^2 = 1,048,576$; $\gamma D_{\max} = 262,144$; $e^{-8} \times 1024 < 1 \Rightarrow$ effective LUT entries = 8 (indices 0–7); $\text{lut}[8] = 0$.

Distance Accumulator & 2-Cycle Pipeline Drain

3-stage pipeline per dimension: subtract $\rightarrow$ square $\rightarrow$ accumulate. After dimension 255, stages 1 and 2 hold δ_{254} , δ_{255} in flight. **Two drain cycles** flush them before the result is valid:

$$D = \sum_{i=0}^{255} (x_i - sv_i)^2, \quad T_{\text{dist}} = 256 \cdot \text{LAT} + 2 \text{ cycles}$$

Without the drain (v6 bug): FSM exits ACCUMULATE at cycle $T+0$; δ_{254} and δ_{255} are never added $\Rightarrow$ accuracy collapse.

Fix (compute_core.sv):

```
logic [1:0] drain_cnt;
if (drain_cnt == 2'd2) next_state = OUTPUT;
```

Verified by tb_dist_zero.sv: all features = sv features = 1.0 ($\mathbb{Q}_{6.10}$ 0x0400) $\Rightarrow$ kernel_out = 1024 ($e^{-\gamma \cdot 0} = 1$). Reading one cycle early gives kernel_out < 1024.

Kernel (Horner + Range-Reduction LUT)

$$K = e^{-\gamma d^2} = e^{-I} \cdot e^{-F}, \quad I = \lfloor \hat{p}/2^{10} \rfloor, \quad F = (\hat{p} \bmod 2^{10})/2^{10}, \quad \hat{p} = \hat{\gamma} \cdot \hat{d}^2/2^{10}$$

LUT: $\text{lut}[I] = \lfloor e^{-I} \cdot 1024 \rfloor$, $I \in [0, 7]$.

Horner polynomial for e^{-F} , $F \in [0, 1)$, effective degree 6:

$$e^{-F} \approx 1 - F + \frac{F^2}{2} - \frac{F^3}{6} + \frac{F^4}{24} - \frac{F^5}{120} + \frac{F^6}{720}$$

RTL defines 15 HORNER states (HORNER_14 down to HORNER_0); coefficients a_7 – a_{14} truncate to zero in $\mathbb{Q}_{6.10}$. FSM runs all 15 steps regardless. Kernel FSM total: $T_{\text{kernel}} = 2 + 15 + 1 = 18$ cycles.

Output: $\hat{K} = (1\text{ut}[I] \cdot \hat{e}^{-F}) \gg 10 \in [0, 1024]$; $\hat{K} = 1024 \Leftrightarrow K = 1$ (when $\mathbf{x} = \mathbf{sv}$).

FSM Cycle Budget (LAT = 3, 500 SVs)

$$T_{\text{beat}} = \underbrace{256 \times 3}_{\text{LOAD}} + 500 \times (\underbrace{256 \times 3 + 2}_{\text{dist}} + \underbrace{18}_{\text{kernel}}) + 1 = 768 + 500 \times 788 + 1 = 394,769 \text{ cycles}$$

At 40 MHz: $394,769 / 40 \times 10^6 \approx \mathbf{9.87 \text{ ms/beat}}$.

Phase	Cycles	What happens
LOAD_INPUT	768	256 features $\times$ LAT=3; GPIO addr, <code>la_data_in[15:0]</code>
COMPUTE_DIST $\times 500$	385,000	SV rows 0–499; 2 drain cycles per SV
COMPUTE_KERNEL $\times 500$	9,000	Horner LUT; LAT-independent
WRITE_CLASS / argmax	~ 1	5-way comparator; <code>class_out[2:0]</code> , <code>IRQ[0]</code>

OpenLane Physical Implementation Flow

Stage	Tool	Key output / note
Synthesis	Yosys	RTL $\rightarrow$ 157,991-cell netlist (sky130_fd_sc_hd)
Floorplan	OpenROAD	Die area, IO pins, 15% utilization target
PDN	OpenROAD	VDD/VSS stripes and rings inserted before placement
Placement	OpenROAD (RePlAce)	Global (min wire length) $\rightarrow$ detailed (legalize)
CTS	TritonCTS	Clock buffer tree; skipped in wrapper (RUN_CTS=0)
Routing	TritonRoute	Global channels $\rightarrow$ detailed metal traces; antennas here
STA	OpenSTA	WNS/TNS; RUN_MCSTA=1: TT/FF/SS corners
DRC	Magic / KLayout	Physical polygon rule checks
LVS	Netgen	Extracted netlist vs. gate-level netlist
GDSII export	Magic / KLayout	Final layout file

Two-stage harden: Core hardened first as black-box macro (job 92840); wrapper harden places macro and routes only SoC-to-macro interface (job 92861).

CTS — Clock Tree Synthesis

CTS inserts buffers/inverters so every flip-flop receives the clock edge at nearly the same time. Without it, a long wire driving 157K flops would have large skew, eating setup and hold margins.

Why RUN_CTS=0 in wrapper: The clock tree already exists inside the hardened `svm_compute_core` macro. Re-running CTS on the wrapper would try to rebuild the tree from scratch while the tool cannot see inside the black-box macro. It would fight the existing internal tree, add skew on top, and likely violate core internal timing. The wrapper clock path is trivially simple: `wb_clk_i` $\rightarrow$ `ICG` $\rightarrow$ `svm_gclk` $\rightarrow$ `one macro`. No tree needed.

Multi-Corner STA (v10, RUN_MCSTA=1)

Corner	Conditions	WNS	Violations	Notes
TT	25°C, 1.80 V	+3.96 ns	0	Nominal
FF	–40°C, 1.95 V	+11.24 ns	0	Fast cells = more slack
SS	100°C, 1.60 V	–14.56 ns	163	Expected for sky130 — non-blocking

Why SS fails setup (not hold): Slow-slow $\rightarrow$ slow transistors at high temp/low voltage $\rightarrow$ combinational paths take longer $\rightarrow$ data arrives late (setup violation). Hold violations appear in FF (cells settle so quickly data may change before hold expires).

Why non-blocking: sky130 SS at 100°C/1.60 V is an unusually harsh corner that even well-designed commercial chips often fail at sky130's nominal clock period. Device operates at room temperature on a stable supply; TT and FF are both clean.

ICG — Integrated Clock Gate

A **latch + AND gate**: latch captures the clock-enable on the falling edge (glitch-free); AND combines it with the clock. This gates `svm_gclk` from `wb_clk_i` while the core is idle.

Why not a plain AND: A combinational AND on a clock line creates glitches — if the enable changes while the clock is high, a partial pulse propagates. The latch ensures the enable is only sampled at the falling edge, so only complete clock cycles pass through.

Power impact: Duty cycle $\approx 1.3\%$ at 80 bpm (LAT=3). $P_{\text{tree}} = 55.25 \times 0.37 \approx 20.4 \text{ mW}$ (37% of active power is clock tree). ICG eliminates this for 98.7% of the time. Idle power $\approx 34.85 \text{ mW}$ (leakage only, clock stopped).

DRC vs. LVS

Check	Full Name	Verifies	Tool	m5 result
DRC	Design Rule Check	Physical polygons obey foundry mfg. rules	Magic/KLayout	Core: 0; Wrapper: 554 antenna (advisory)
LVS	Layout vs. Schematic	Extracted netlist = gate-level netlist	Netgen	Core: 0; Wrapper: 1,683 boundary artifacts

Antenna violations (554 net / 808 pin): Charge accumulation during plasma etching on long metal runs not yet connected to a gate. Advisory, not blocking. Mitigated by antenna diodes or metal jumpers in production.

LVS 1,683 errors on wrapper: Macro interface boundary artifacts — the extractor sees hardened core ports as unresolved black-box connections. Not logical errors. Core LVS is clean (0 errors).

Caravel SoC Harness

Caravel is the fixed SoC frame for Google-sponsored sky130 shuttle runs. You design `user_project_wrapper`; Caravel provides:

Component	Details
RISC-V (PicoRV32)	Management processor; runs C firmware from on-chip flash; issues Wishbone writes
Wishbone B4	Management core = master; user project = slave at base <code>0x3000_0000</code>
GPIO mux	38 configurable pads; SRAM addr GPIO[28:10], read strobe GPIO[29]
Logic Analyzer	128 <code>la_data_in/out</code> wires; SRAM read data on <code>la_data_in[15:0]</code>
Clock/reset	<code>wb_clk_i</code> , <code>wb_rst_i</code> routed to wrapper boundary

Wishbone base address 0x3000_0000: Set by Caravel’s hardwired address decoder, not your RTL. Your design only sees lower offset bits. Changing it requires modifying Caravel’s management core — not permitted.

wb_ack_o: Slave handshake signal. The master stalls (`wbs_stb_i` held high) until `wb_ack_o` is asserted. If the slave never asserts it, the bus hangs permanently — the master cannot time out, the MCU deadlocks, and no further Wishbone traffic is possible until reset.

Offset	Register	Dir	Notes
+0x04	CONTROL	RW	[0]=start, [1]=vbatt_ok, [2]=vbatt_warn
+0x08	STATUS	RO	[8:6]=class, [9]=sample_rdy, [0]=done
+0x0C	NUM.SAMPLES	RW	[9:0] beats (resets to 0; fix: default 1000)
+0x10--0x20	NUM.SV[0–4]	RW	[7:0] SVs/class (8-bit, max 255; total ≤ 500)
+0x24	PARAM.WR	WO	[19]=en, [18:16]=addr, [15:0]=Q6.10
+0x28	ALPHA.WR	WO	[24:16]=sv_idx (9-bit), [15:0]=alpha Q6.10

RR Normalization & Feature Vector

Feature vector (256 dims): 128 single-beat morphology (± 64 samples, amplitude-normalized) + 64 10-beat mean template + 64 RR-interval history (99 intervals $\rightarrow$ 64 pts).

RR normalization: Raw RR intervals divided by `NORMAL_RR`=308 ms (median NSR, MIT-BIH+SVDB+INCART) expressed as a ratio, making rhythm features rate-independent across patients. The 64 RR dims are the rhythm-slice dimensionality.

Q6.10 Quantization & Optimal SV Allocation

3-way comparison (`confusion_3way.png`): sklearn joint OVR (97.67%) $\rightarrow$ ASIC binary OVR float (98.33%) $\rightarrow$ ASIC Q6.10. Separates *model architecture* effect from *quantization* effect. Binary OVR model *outperforms* sklearn in float; equal final accuracy is coincidental.

Why low- $|\alpha|$ tail SVs hurt Q6.10: Each kernel evaluation contributes a quantization error ε_j to the score sum. High- $|\alpha_j|$ SVs: large signal, noise ratio good. Low- $|\alpha_j|$ tail SVs: near-zero signal, same $|\varepsilon_j| \Rightarrow$ noise-to-signal ratio inverts. Accumulated tail noise can flip boundary samples.

SV count sweep (`sv_sweep.py`), uniform N per class:

N /class	Total	Float	Q6.10	Flips	Note
90	450	98.00%	98.00%	0	
100	500	98.33%	98.00%	1	HW ceiling (equal split)
120	600	98.67%	98.67%	0	global optimum
150	750	98.00%	98.00%	0	tail noise degrades

Global optimum at $N = 120$ /class (600 total) exceeds hardware ceiling. Float accuracy peaks then *falls* above $N = 120$ — low- $|\alpha|$ tail SVs corrupt the boundary in float too.

Recommended (within 500-SV hardware): [95, 95, 95, 120, 95] — VT at per-class optimum, total = 500.

Allocation	Float	Q6.10	Flips
Baseline [100,100,100,100,100]	98.33%	98.00%	1 (Normal $\rightarrow$ SVT)
Optimal [95,95,95,120,95]	98.33%	98.33%	0

Next iteration (v11 reharden): `alpha_table[500]` $\rightarrow$ [600], `alpha_addr` 9-bit $\rightarrow$ 10-bit, `NUM_SAMPLES` reset default 0 $\rightarrow$ 1000. Target: 98.67% Q6.10, 0 flips. SRAM has room; only on-chip RTL changes required.